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Effect of orientation on the drag of a cone settling in a Bingham plastic fluid

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ABSTRACT

Effects of the orientation and apex angle on the settling velocity of conical particles in Bingham plastic fluids are numerically investigated over a Reynolds number range of $1 \le Re \le 100$, Bingham number range of $0 \le Bn \le 100$, and cone angle range of $20^\circ \le \alpha \le 150^\circ$. Governing equations (of continuity and momentum) are solved numerically using the finite element method to obtain velocity and pressure fields that are postprocessed to obtain values of the drag coefficient. Furthermore, the effect of cone orientation on the flow field is visualized and explored in terms of streamline contours and the morphology of yielded/unyielded regions. Finally, the obtained values of drag are correlated via a simple predictive expression in terms of the modified Reynolds number. The effect of the cone angle on drag is found to be moderate because the drag force acting on the lateral surface is a small fraction of the overall drag.

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Introduction

The settling behavior of variously shaped objects in viscoplastic fluids has been researched owing to fluid viscoplasticity in several practical applications; e.g., the processing of concentrated suspensions, pastes, and foams. Specific examples of research are the formulation of processed foodstuffs and pharmaceutical products, and the design of slurry pipelines for the transportation of mineral slurries and mine tailings in flumes, launders, and open channels. Depending on the envisaged application, it is sometimes desirable to keep particles in suspension to ensure the homogeneity of such products. There are occasions when it is desired to effect solid–liquid separation, such as in the dewatering of coal–water and other mineral slurries. Additional potential applications of such single-particle studies are found in the modeling of fluidized beds and fixed-bed reactors where clusters of particles that may be homogeneous or heterogeneous in nature are encountered (e.g., Ghadirian & Arastoopour, 2016; Nikolopoulos, Papafotiou, Nikolopoulos, Grammelis, & Kakaras, 2010). The need to estimate the settling velocity of a particle in such fluids thus frequently arises in process design calculations. This information is typically expressed using dimensionless parameters, such as the drag coefficient and Reynolds number, for a given shape and class of fluids, such as Newtonian and non-Newtonian fluids (Clift, Grace, & Weber, 1978). Consequently, reliable expressions for the prediction of drag as a function of the pertinent governing parameters of variously shaped objects (Chhabra, Agarwal, & Sinha, 1999; Clift et al., 1978; Rajitha, Chhabra, Sabiri, & Comiti, 2006) have appeared in the literature for Newtonian and non-Newtonian fluids. Indeed, the modified forms of such drag expressions are also used to approximate the interphase momentum transfer in concentrated systems, such as circulating fluidized beds and packed beds (Ghadirian & Arastoopour, 2016; Nikolopoulos et al., 2010). Because each nonspherical particle is unique in shape, such studies continue unabated even for Newtonian fluids (Wang, Qi, & You, 2009; Wang, Qi, & Zhu, 2011). Although a combination of numerical and experimental studies has provided satisfactory predictive methods for drag on nonspherical particles as far as simple Newtonian fluids are concerned (Chhabra, 2006), a quick glance of the pertinent literature clearly shows that little information is available on the drag behavior of cones, even in Newtonian fluids (Sharma & Chhabra, 1991). The motion of cones in soft solids, such as butter, jams, and jellies, and other pasty materials is frequently used to measure the yield strength or spreadability in quantifying quality in food, pharmaceutical, and personal care product sectors. Additional applications relate to soil testing and the design of farming tools, such as ploughs and spades (Durban & Fleck, 1992; Koumoto & Houlsby, 2001). Interestingly,

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Nomenclature

Bn	Bingham number $\left(=\tau_0 D/(\mu_B U_\infty)\right)$, dimensionless
Bn^*	Bingham number $\left(=\tau_0/\left(K\left(U_\infty/D\right)^n\right)\right)$, dimensionless
C_D	Drag coefficient, dimensionless
C_D^*	Drag coefficient $\left(=4F_D/\left(\tau_0\pi D^2\right)\right)$, dimensionless
C_{DF}	Friction drag coefficient, dimensionless
C_{DP}	Pressure drag coefficient, dimensionless
C_P	Pressure coefficient, dimensionless
D	Diameter of the cone, m
D_∞	Size of the computational domain, dimensionless
F_D	Drag force exerted on the cone, N
F_{DP}	Pressure component of the drag force on the cone, N
K	Flow consistency index, Pa s^n
m	Growth rate parameter, dimensionless
n	Flow behavior index, dimensionless
p	Pressure, dimensionless
r, z	Cylindrical co-ordinates, m
Re	Reynolds number $\left(\equiv \rho D U_\infty/\mu_B\right)$, dimensionless
Re^*	Modified Reynolds number, Eq. (17), dimensionless
$\mathbf{U}$	Velocity vector, dimensionless
U_∞	Settling velocity, m/s
U_z	Velocity component in the z-direction, dimensionless
U_r	Velocity component in the r-direction, dimensionless

Greeks

α	Apex angle of the cone, $^\circ$
$\dot{\boldsymbol{\gamma}}$	Rate of the strain tensor, dimensionless
$\dot{\boldsymbol{\gamma}}_e$	Regularization parameter in the Bercovier and Engelman model, dimensionless
η	Apparent viscosity, dimensionless
μ_B	Plastic viscosity, Pa s
μ_y	Yielding viscosity, Pa s
ρ	Density of the fluid, kg/m^3
$\boldsymbol{\tau}$	Extra stress tensor, dimensionless
τ_0	Yield stress, Pa
∇	Del operator, m^{-1}

Subscripts

∞	Inlet condition
s	Cone surface condition

most of these substances also exhibit so-called viscoplastic behavior (Chhabra & Richardson, 2008). Aside from the overwhelming pragmatic relevance of the described geometry, such model studies also contribute to the overall advance of the sub-field of particulate flows. The present study thus focuses on flow over a cone oriented with its apex upstream or downstream with reference to the oncoming stream of a viscoplastic fluid. However, prior to undertaking a detailed discussion of the new results obtained in this work, it is instructive to briefly describe pertinent studies available in the literature on nonspherical particles in non-Newtonian fluids in general and on cones in particular.

Previous works

As far as the prediction of drag on cones in Newtonian fluids is concerned, the scant literature up to 1978 was thoroughly reviewed by Clift et al. (1978). These studies were subsequently revisited by others (Chhabra et al., 1999; Ganser, 1993). Sharma and Chhabra (1991) and Borah and Chhabra (2005) reported experimental results for wall effects and the drag exerted on freely falling cones in Newtonian and power-law fluids. The work of Borah and Chhabra (2005) spanned the range of flow behavior index of power-law fluids $0.4 \le n \le 1$ and the range of Reynolds numbers $0.019 \le Re \le 507$. The results of Borah and Chhabra (2005) for Newtonian fluids were found to be consistent with the predictions of Ganser (1993). However, they developed empirical correlations for both wall effects and drag coefficients for both orientations of cones applicable to Newtonian and power-law fluids. A recent study by Yu, Pan, Lin, & Shao (2014) numerically examined the orientation of a cone settling in Newtonian fluids in a vertical duct. Depending on the apex angle, the cone was observed to settle with its apex upward or downward. At $Re = 30$ and for cone angles smaller than 48°, irrespective of the initial orientation, the cone settled with its apex pointing upward; i.e., presenting the maximum projected area normal to the direction of sedimentation at the cone's first contact with the fluid. Meanwhile, for an apex angle exceeding 52°, the cone settled with its apex pointing downwards. These predictions are in line with their experimental results. However, all these observations pertain to a cone diameter/duct ratio of about 0.25 and these trends may well depend on this ratio. It is thus far from straight forward to predict the trends for an unconfined case and/or for other values of the blockage ratio. As far as known to us, no other numerical results are available for cones in Newtonian or non-Newtonian fluids.

In contrast, experimental results on cylinders, spheres, spheroids, hemispheres, and disks in viscoplastic materials are available in the literature (Chhabra, 2006), though the numerical predictions of drag for spheres (Nirmalkar, Chhabra, & Poole, 2013), spheroids (Gupta & Chhabra, 2014), and hemispheres (Nalluri, Patel, & Chhabra, 2015; Patel, Patel, Raja, & Chhabra, 2015) at finite Reynolds numbers in Bingham plastic fluids have been reported very recently. For a given shape and orientation, the hydrodynamic drag force is stronger than that in Newtonian fluids because of the fluid yield stress. While the quantitative effect of the fluid yield stress varies from one shape to another, this trend is qualitatively similar across all shapes studied so far. To the best of our knowledge, only Ahonguio, Jossic, & Magnin (2015) have studied the flow of yield stress fluids passing conical objects in the creeping-flow regime. They concluded that the drag coefficient depends directly on the yield stress and cone apex angle. They also expressed the drag coefficient as a function of the yield stress and cone apex angle. This work also demonstrates that the relative contribution of the drag force stemming from the lateral surface of the cone progressively diminishes with increasing cone angle. It is thus fair to conclude from the preceding discussion that no prior results are available on the motion of cones in viscoplastic fluids, especially at finite Reynolds numbers. The present work fills this gap in the literature by supplementing the results of Ahonguio et al. (2015). In particular, continuity and momentum equations are solved numerically for the steady flow of Bingham plastic fluids over cones with their apex oriented upstream and downstream of the free stream. Extensive results are presented here on the morphology of yielded/unyielded regions and drag coefficient at Reynolds numbers ranging $1 \le Re \le 100$, Bingham numbers ranging $0 \le Bn \le 100$, and cone angles ranging $20^\circ \le \alpha \le 150^\circ$. The paper is concluded

by making comparisons with experimental results taken from the literature.

Problem formulation

Consider the flow of a uniform free stream (with velocity U_∞) of a Bingham plastic liquid over a cone (of diameter D) with its apex pointing upward or downward with reference to the direction of mean flow, as shown in Fig. 1. The flow is assumed to be axisymmetric. Within the framework of these assumptions and considering incompressible, steady, and laminar flow, the velocity and pressure fields are governed by the continuity and momentum equations written in their dimensionless forms as:

Continuity equation:

$$\nabla \cdot \mathbf{U} = 0, \tag{1}$$

Momentum equation:

$$(\mathbf{U} \cdot \nabla)\mathbf{U} = -\nabla p + \frac{1}{Re}\nabla \cdot \boldsymbol{\tau}. \tag{2}$$

The extra-stress tensor for an incompressible fluid is given by,

$$\boldsymbol{\tau} = 2\eta\dot{\boldsymbol{\gamma}}. \tag{3}$$

For a Bingham plastic fluid, the constitutive relation of the extra-stress tensor $\boldsymbol{\tau}$ is given by (Macosko, 1994),

$$\begin{aligned} \boldsymbol{\tau} &= \left(1 + \frac{Bn}{|\dot{\boldsymbol{\gamma}}|}\right)\dot{\boldsymbol{\gamma}} \quad \text{if} |\boldsymbol{\tau}| > Bn; \\ \dot{\boldsymbol{\gamma}} &= 0, \qquad\qquad\quad \text{if} |\boldsymbol{\tau}| \le Bn. \end{aligned} \tag{4}$$

The Bingham number, $Bn = (\tau_0 D)/(\mu_B U_\infty)$, is the nondimensional yield stress of the fluid. The problem is closed by prescribing physically realistic boundary conditions as follows. The usual no-slip condition is used on the surface of the cone. Symmetry conditions are applied on this plane because the flow is expected to be symmetric about the vertical centerline; i.e., $U_r = 0$ and $\partial U_z/\partial r = 0$. For a truly unconfined flow, the far away boundary condition is given by $r \to \infty$, $U_z = 1$, and $U_r = 0$. Naturally, it is not possible to simulate a truly unconfined flow in a numerical study. This approximation is made here by enclosing the cone in a large spherical envelope of fluid with diameter D_∞, as discussed in more detail in a later section. The lower half of the envelope is treated as the inlet where the conditions $U_z = 1$ and $U_r = 0$ are adopted. On the upper half (i.e., the exit surface), the standard outflow condition is used (i.e., $\partial U_r/\partial z = 0$ and $\partial U_z/\partial z = 0$), though gradients in the lateral direction can still exist.

The preceding governing equations and boundary conditions have been scaled using U_∞ and D as the characteristic velocity and length scales. The stress has thus been scaled by $\left(\mu_B U_\infty/D\right)$, the pressure by ρU_∞^2, and the viscosity by μ_B. The inherently discontinuous nature of the Bingham model makes direct implementation in a numerical scheme far from straightforward. Strictly speaking, one must solve the governing equations using a method such as the operator splitting method (Huilgol & Kefayati, 2015), augmented Lagrangian method or another method based on variational inequalities (Glowinski, 1984) in mapping both yielded and unyielded regions. Because the yield surfaces are not known a priori, this approach is cumbersome and is highly computational-resource intensive. Many authors have thus devised regularized forms of the Bingham plastic equation that convert the abrupt transition between solid-like and fluid-like regions into a gradual transition (Glowinski & Wachs, 2011). Three such schemes, namely exponential (Papanastasiou, 1987), biviscous (O'Donovan & Tanner, 1984), and Bercovier and Engelman (1980) schemes, have gained wide acceptance in the literature and they are used in the present work. The scalar viscosity functions, η, are given as follows.

Papanastasiou regularization

In the scheme of Papanastasiou,

$$\eta = 1 + \frac{Bn\,[1 - \exp(-m|\dot{\gamma}|)]}{|\dot{\gamma}|} \tag{5}$$

Evidently, $m \to \infty$ relates to the true Bingham behavior. Large values of m must therefore be used in Eq. (5) to approach the behavior of a true Bingham fluid.

Biviscous approximation

In the biviscous framework, the unyielded material is assigned a high viscosity (yielding viscosity, μ_y) that is often several orders of magnitude greater than the Bingham viscosity (μ_B). This model is written as:

$$\begin{aligned} \eta &= \frac{\mu_y}{\mu_B}, \text{ for } \tau \le Bn; \\ \eta &= 1 + Bn\left[1 - \left(\frac{1}{(\mu_y/\mu_B)}\right)\right]\frac{1}{\dot{\gamma}}, \text{ for } \tau > Bn. \end{aligned} \tag{6}$$

Naturally, $\mu_y \to \infty$ corresponds to the true Bingham behavior and hence large values of μ_y must be employed.

Bercovier–Engelman approximation

Strictly speaking, a Bingham material deforms like an elastic solid for $\tau \le Bn$; i.e., $\dot{\gamma} = 0$. This condition is approached here by allowing a very low strain rate in the unyielded region, and the Bingham model is written as:

$$\eta = 1 + Bn\left(\frac{1}{\dot{\gamma} + \dot{\gamma}_e}\right). \tag{7}$$

In this case, values of $\dot{\gamma}_e$ as small as possible must be used to mimic the true Bingham behavior.

The success of the three regularization schemes hinges on a suitable choice of the single regularization parameter (m or μ_y or $\dot{\gamma}_e$). All the schemes have been shown to yield results for a range of geometries that are virtually indistinguishable from each other and the predictions are also in excellent agreement with those obtained using the operator-splitting and augmented Lagrangian approach in cases for which reliable results are available, as recently discussed in detail (Baranwal & Chhabra, 2017). The present work used the exponential approach described by Eq. (5) (Papanastasiou, 1987) to obtain most of the results, while limited simulations were carried out using the other two approaches to substantiate the new results reported herein.

The rate-of-strain tensor, $\dot{\boldsymbol{\gamma}}$, is given by,

$$\dot{\boldsymbol{\gamma}} = \frac{1}{2}(\nabla\mathbf{U} + \nabla\mathbf{U}^T). \tag{8}$$

The magnitudes of $\boldsymbol{\tau}$ and $\dot{\boldsymbol{\gamma}}$ tensors are given by,

$$|\boldsymbol{\tau}| = \sqrt{\frac{1}{2}\mathrm{tr}(\boldsymbol{\tau}^2)}, \tag{9}$$

$$|\dot{\boldsymbol{\gamma}}| = \sqrt{\frac{1}{2}\mathrm{tr}(\dot{\boldsymbol{\gamma}}^2)}. \tag{10}$$

Dimensional considerations applied to the governing equations and boundary conditions suggest that, for a given apex angle and orientation, the momentum transfer characteristics are characterized by the Reynolds number (Re) and Bingham number (Bn) defined as:

$$Re = \frac{\rho D U_\infty}{\mu_B}, \tag{11}$$

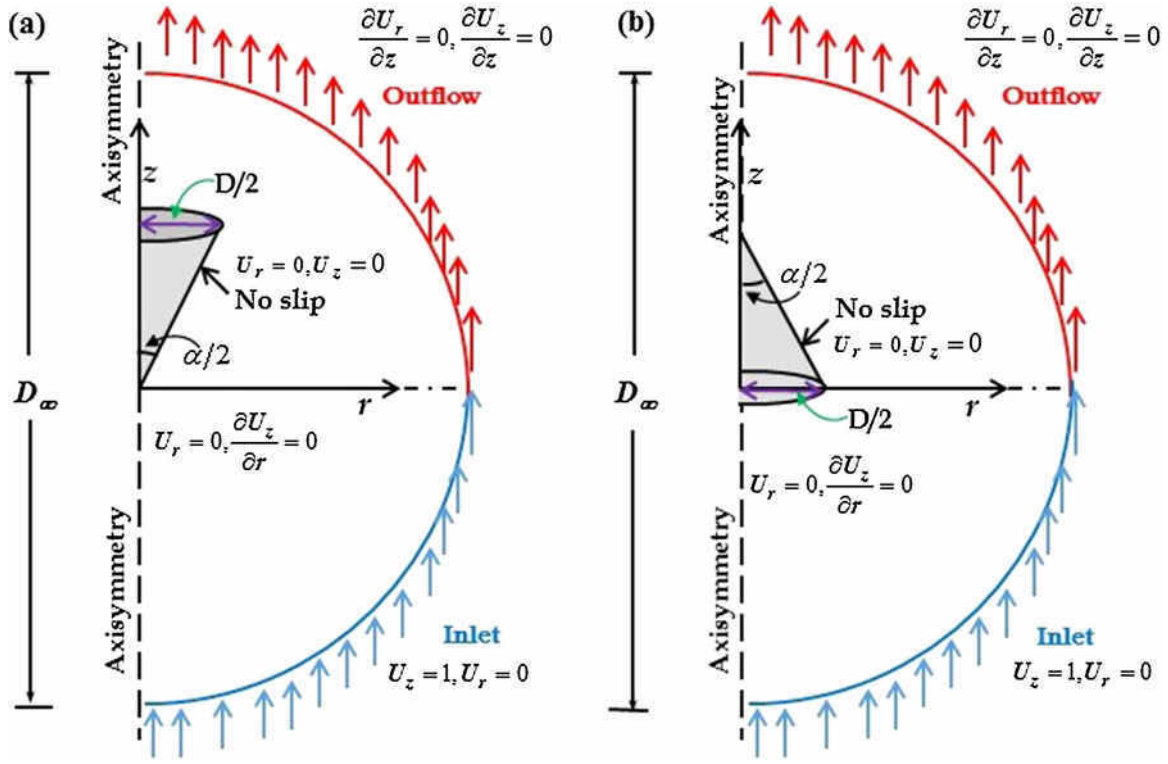


Fig. 1. Schematics of the physical domain: (a) configuration 1 and (b) configuration 2.

$$Bn = \frac{\tau_0 D}{\mu_B U_\infty}. \tag{12}$$

The numerical solution of the momentum and continuity equations maps the flow domain in terms of primitive variables (u, v, p), which are then postprocessed (via integration of the normal and shearing forces acting on the surface of the cone) to deduce the drag coefficients (Nirmalkar et al., 2013):

$$C_{DF} = \frac{8F_{DF}}{\rho U_\infty^2 \pi D^2}, \tag{13}$$

$$C_{DP} = \frac{8F_{DP}}{\rho U_\infty^2 \pi D^2}, \tag{14}$$

$$C_D = C_{DF} + C_{DP}, \tag{15}$$

$$C_p = \frac{2(P_S - P_0)}{\rho U_\infty^2}. \tag{16}$$

In summary, C_{DF}, C_{DP}, and C_D are expected to be functions of the Reynolds number, Re, and Bingham number, Bn, for a given orientation and apex angle. The present work explores and develops these relationships.

Numerical methodology and choice of computational parameters

The aforementioned governing differential equations subject to the boundary conditions were solved numerically using the finite element based solver, COMSOL Multiphysics (Version 4.3a). The nonuniform grid structure employing both triangular and quadrilateral cells was used for meshing the computational domain. Fine meshes were used near the cone and yield surfaces. The resulting system of equations was solved with the steady, axisymmetric, laminar flow module using the PARDISO solver with a parametric sweep. The viscosity of the fluid was input through a user-defined function. A relative convergence criterion of 10^{-5} was found to be adequate for both the continuity and momentum equations, for no change in results was observed on further reduction of this value. Within this framework, the individual drag and total drag stabilized at least up to four significant digits. Similarly, a von Mises yielding criterion (relative) of 10^{-5} was employed to delineate yield surfaces (Macosko, 1994).

Table 1
Grid independence test for $\alpha = 60^\circ$ at $Re = 100$ and $m = 10^5$ for configuration 1 and configuration 2.

Grid	Np	Elements	$Bn = 0$		$Bn = 100$	
			C_D	C_{DP}	C_D	C_{DP}
Configuration 1 (apex downward)						
G1	350	72,906	1.0391	0.4571	28.380	22.928
G2	450	89,384	1.0402	0.4573	28.394	22.929
G3	550	112,712	1.0405	0.4573	28.397	22.930
Configuration 2 (apex upward)						
G1	140	37,975	1.2880	1.0604	28.517	23.440
G2	180	47,623	1.2884	1.0610	28.522	23.443
G3	220	56,448	1.2887	1.0612	28.527	23.445

N_p = number of elements on the surface of the cone.

Following the strategy employed in some of our recent studies (Nirmalkar et al., 2013; Nalluri et al., 2015), an appropriate value of D_∞ and numerical mesh were selected in a series of numerical experiments. The value of (D_∞/D) was systematically varied as $D_\infty/D = 300$, 400, and 600 and the resulting values of drag were monitored at $Re = 1$ for the two extreme values of the Bingham number (i.e., $Bn = 0$ and $Bn = 100$). For both configurations, the results obtained with $D_\infty/D = 400$ and $D_\infty/D = 600$ were virtually indistinguishable from each other and $D_\infty/D = 400$ was thus used in the present work.

Similarly, three computational grids (i.e., G1, G2, and G3, detailed in Table 1), differing in terms of the number of nodes on the surface of the cone and in the entire computational domain, were created to select an appropriate grid. Table 1 summarizes the grid effect on the results for the maximum value of $Re = 100$ used

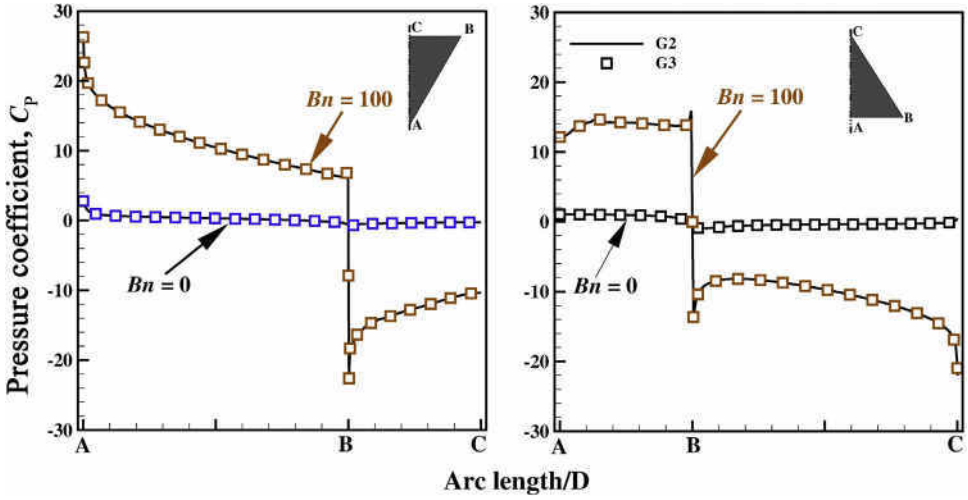


Fig. 2. Effect of the grid on the pressure coefficient at $Re = 100$, $\alpha = 60^\circ$, and $Bn = 0$ and $Bn = 100$, for configuration 1 and 2.

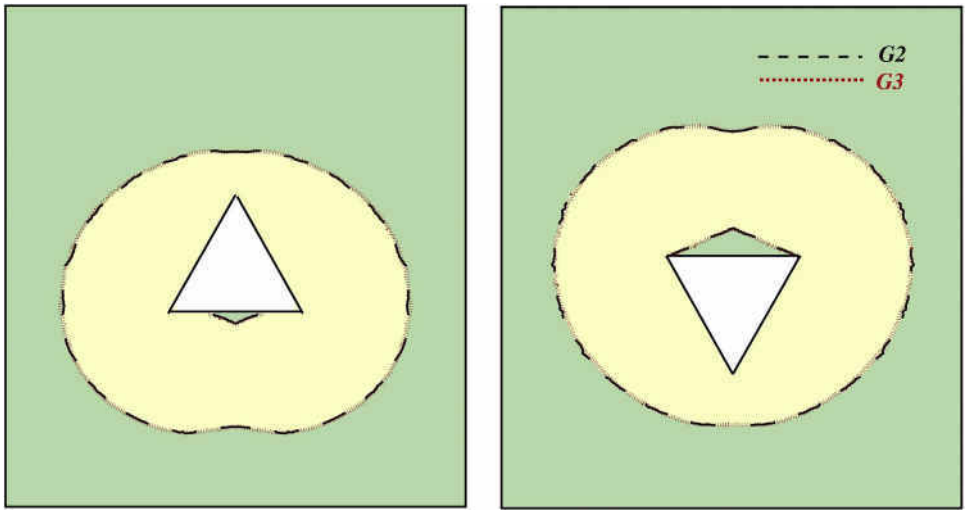


Fig. 3. Effect of grid on the structure of the yielded/unyielded region for a cone with an apex angle 60° at $Re = 100$ and $Bn = 100$.

Table 2
Effect of the growth rate parameter, m, on pressure and total drag coefficient at $\alpha = 60^\circ$ and $Re = 100$ for configuration 1 and configuration 2.

m	$Bn = 10$		$Bn = 100$	
	C_D	C_{DP}	C_D	C_{DP}
Configuration 1 (apex downward)				
10^4	4.2379	3.0731	28.390	22.926
10^5	4.2384	3.0737	28.394	22.929
10^6	4.2388	3.0739	28.401	22.930
Configuration 2 (apex upward)				
10^4	4.4812	3.9893	28.512	23.439
10^5	4.4856	3.9899	28.522	23.443
10^6	4.4860	3.9901	28.531	23.445

here and for the two extreme values of the Bingham number, i.e., $Bn = 0$ and $Bn = 100$. The values are seen to change very little in going from G2 to G3. Furthermore, Figs. 2 and 3 respectively show grid effects in terms of the surface pressure distribution and yield surfaces, where the predictions of G2 and G3 are seen to be virtually indistinguishable from each other. Thus, on either count, grid G2 is adequate for resolving the thin boundary layers.

Next, Table 2 summarizes the results obtained with three different values of m while Fig. 4 contrasts the predictions of the three regularization methods in terms of the yield surfaces. The three predictions are seen to be in close agreement. Minor differences in the far away yield surfaces had little effect on the velocity gradients at the surface of the cone; see Table 3. The present work therefore employed the exponential regularization, described by Eq. (5), with $m = 10^5$. Finally, to check the validity of the steady-state assumption, limited time-dependent (full-domain) simulations were conducted at the maximum value of the Reynolds

Table 3
Effect of the regularization model on the pressure and total drag coefficient at $\alpha = 60^\circ$ and $Re = 100$, for configuration 1 and 2.

Regularization model	$Bn = 10$		$Bn = 100$	
	C_D	C_{DP}	C_D	C_{DP}
Configuration 1 (apex downward)				
Bi-viscosity model $\left(\mu_y/\mu_B = 10^5\right)$	4.2383	3.0644	28.389	22.974
Papanastasiou model $\left(m = 10^5\right)$	4.2384	3.0737	28.394	22.929
Bercovier-Engleman model $\left(\dot{\gamma}_e = 10^{-10}\right)$	4.2386	3.0647	28.391	22.976
Configuration 2 (apex upward)				
Bi-viscosity model $\left(\mu_y/\mu_B = 10^5\right)$	4.4856	3.9898	28.519	23.431
Papanastasiou model $\left(m = 10^5\right)$	4.4856	3.9899	28.522	23.443
Bercovier-Engleman model $\left(\dot{\gamma}_e = 10^{-10}\right)$	4.4855	3.9897	28.512	23.433

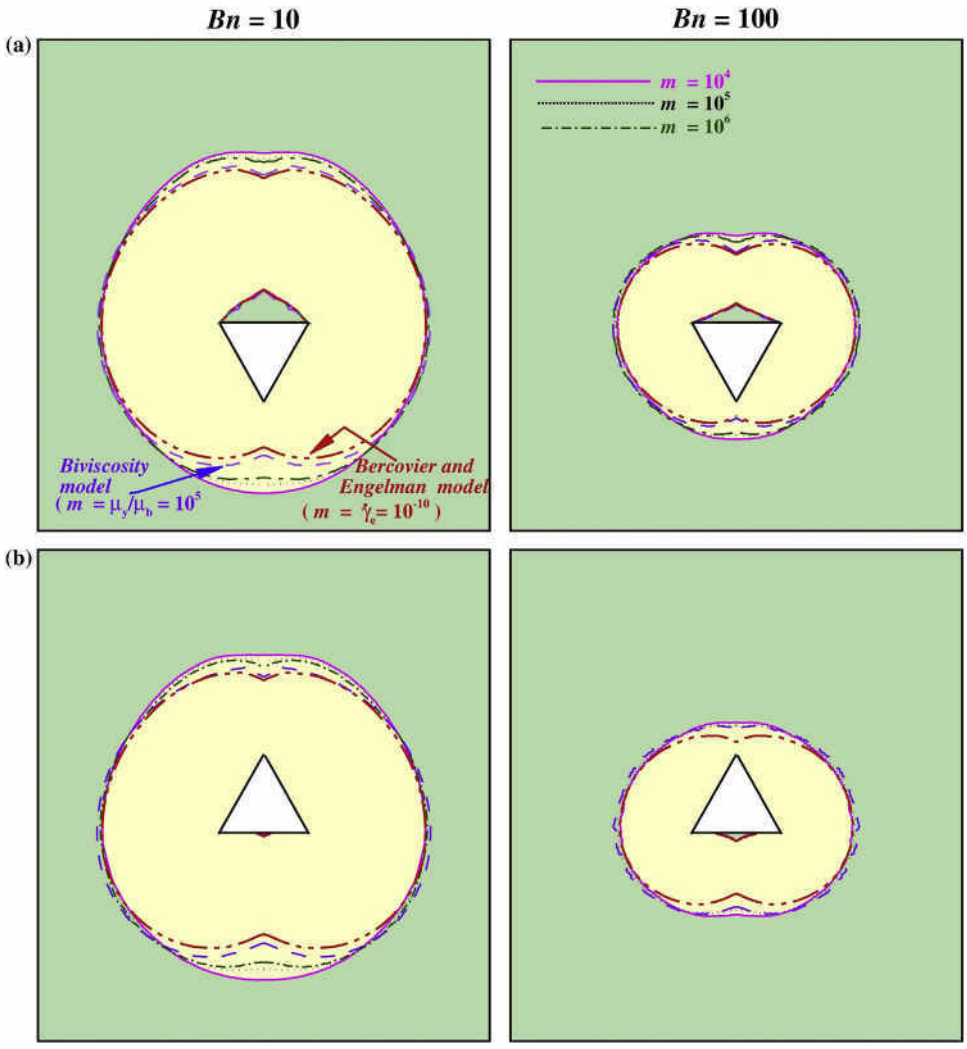


Fig. 4. Effect of the growth rate parameter m on yield surfaces for a cone with an apex angle 60° at $Re = 100$ and $Bn = 10$ and $Bn = 100$ for (a) configuration 1 and (b) configuration 2.

number (i.e., $Re = 100$) and $Bn = 0$, for the presence of yield stress, howsoever small, tends to stabilize the flow as seen in the context of the flow over a spheroid (Gupta & Chhabra, 2014). Indeed, velocity fields were found to be steady over the range of conditions. The adequacy of these numerical parameters was further demonstrated by benchmark comparisons presented in the next section.

Results and discussion

The present study numerically investigated steady axisymmetric flow passing a cone with its flat base oriented upward or downward in a free stream of Bingham plastic fluid at $1 \le Re \le 100$, $0 \le Bn \le 100$, and $20^\circ \le \alpha \le 150^\circ$. The range of the Bingham number is such that it includes the Newtonian fluid behavior ($Bn = 0$) at one end and highly viscoplastic flow at the other end. However, prior to presenting a detailed discussion of the new results, it is worthwhile ascertaining the reliability and precision of the results by way of presenting benchmark comparisons.

Validation of results

General validations were first made for the laminar flow of Newtonian and power-law fluids in a lid-driven square cavity and the present values of the centerline velocities were found to be within ±1% of the literature values (Bell & Surana, 1994; Neofytou, 2005). Similarly, while there are no numerical results available for a cone even in Newtonian fluids, some experimental results (Borah & Chhabra, 2005; Sharma & Chhabra, 1991) are available in the literature, though most of these data relate to $Re > 100$. However, Table 4 contrasts the present predictions with the experimental results of Sharma and Chhabra (1991) and Borah and Chhabra (2005); the two

Table 4
Comparison of the predicted drag coefficient (C_D) at $\alpha = 60^\circ$ fo Newtonian fluids with the experimental results of Borah and Chhabra (2005) and Sharma and Chhabra (1991) in the range of $Re \le 100$.

Re	Present	Borah and Chhabra (2005)	Sharma and Chhabra (1991)
1	24.370	21.759	21.091
2	13.268	12.270	11.764
5	6.3643	6.2086	5.9483
10	3.9923	3.9621	3.8640
15	3.0192	3.1297	3.1114
20	2.5975	2.6778	2.7094
30	2.0926	2.1818	2.2740
40	1.8253	1.9049	2.0338
50	1.6586	1.7230	1.8770
60	1.5445	1.5920	1.7644
70	1.4616	1.4919	1.6785
80	1.3985	1.4121	1.6101
90	1.3483	1.3466	1.5538
100	1.3069	1.2914	1.5065

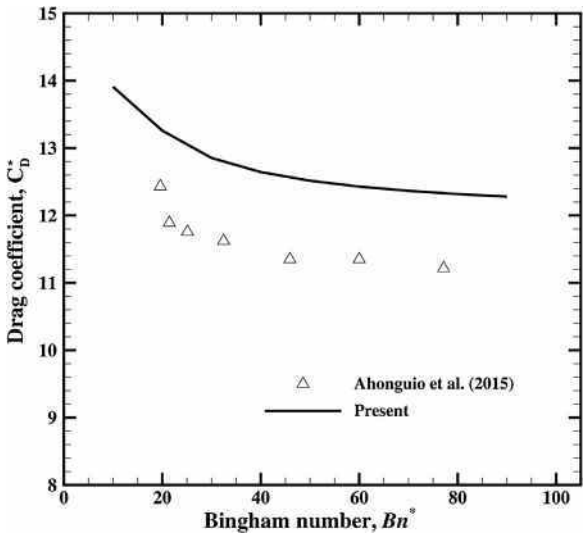


Fig. 5. Comparison of the present predictions with the experimental results of Ahonguio et al. (2015) for Herschel–Bulkley fluids in the creeping-flow regime at $\alpha = 60^\circ$ and $n = 0.37$.

values are seen to be within ±10% of each other in the overlapping range of conditions; i.e., $Re \leq 100$.

Fig. 5 similarly compares the present results with experimental results of Ahonguio et al. (2015) for Herschel–Bulkley fluids ($\tau_0 = 81.5$ Pa, $n = 0.37$, and the consistency $K = 44$ Pa s^n) in the creeping-flow regime. The present values are seen to be about 9% higher than those of Ahonguio et al. (2015), which is well within the accuracy of their experimental results, which is expected to be >10%. Although the present predictions are always (∼10%) higher, they well capture the form of the functional relationship. According to the preceding comparisons with previous numerical and experimental studies coupled with our past experience, the new numerical results are expected to be reliable to within 1%–2%.

Streamline contours and morphology of yielded/unyielded regions

Figs. 6 and 7 show the streamline patterns in Newtonian fluids ($Bn = 0$) for three representative values of the Reynolds number at two orientations of the cone. At low Reynolds numbers (e.g., $Re = 1$), the fluid inertia is small and a fluid element is thus able to follow the body shape without incurring any loss in momentum. However, the inertial forces increase with the Reynolds number. At a critical value of the Reynolds number, an adverse pressure gradient is established at a point on the surface of the cone, and the fluid detaches from the surface of the cone. This occurs, for instance, at $Re \sim 22$–24 for a sphere and $Re \sim 4$–5 for a circular cylinder in cross flow (Clift et al., 1978). Broadly speaking, the critical Reynolds number is lower when the bluff body is blunter. In this work, additional effects arise depending on the orientation and apex angle of the cone. Fig. 6 shows that, for configuration 1, i.e., the fluid approaching the apex of the cone first, there is a symmetric toroidally shaped recirculating region attached to the rear of the cone. This region grows in the flow direction with the Reynolds number and the wake width is of course fixed as the points of separation are fixed in this case. For Newtonian fluids, this transition occurs at about $Re = 0.75$. Meanwhile, for the opposite orientation 2 (Fig. 7), it appears intuitive that not only should this transition occur at lower Reynolds numbers but also the wake need not extend all the way to the tip of the cone. However, as the Reynolds number increases further, the wake does extend beyond the tip of the cone. The critical values of the Reynolds number denoting the onset of the flow separation are summarized in Table 5. Owing to the fluid yield stress, this transition is deferred to higher values of the Reynolds number; see Figs. 6 and 7. Additionally, because a distinguishing feature of Bingham plastic fluids is the coexistence of yielded and unyielded regions, Figs. 8 and 9 show representative results on the morphology of the yield surfaces separating the two regions. Broadly speaking, there are two unyielded regions, a small blob of material adhering at the front stagnation point in the case of a cone oriented with its flat base towards the oncoming fluid stream and the far away body of the unsheared material moving at the velocity of the free stream. Naturally, the increasing Bingham number promotes the growth of the unyielded regions whereas this tendency is countered by the increased Reynolds number. In both orientations, there is a unsheared body of material at the center of the base of the cone due to low stress levels there. All these trends are borne out by the results shown in Figs. 8 and 9 for the two orientations. The fluid-like cavity is seen to extend to about 1 to 3D in the lateral direction and it is slightly longer in the axial direction.

Table 5
Critical Reynolds number for the onset of flow separation at $\alpha = 60^\circ$.

Configuration 1		Configuration 2	
Bn	*Re*	*Bn*	*Re*
0	0.75	0	34
1	0.80	1	62
10	1	2	93
50	20	2.4	98
100	43	5	–

Drag coefficients

Owing to the normal and shearing stresses exerted by the fluid on the surface of the cone, there is a net hydrodynamic force exerted by the fluid in the z-direction that is expressed as the familiar drag coefficient made up of the pressure drag (C_{DP}) and frictional drag (C_{DF}) components. As shown in Fig. 10, in yield stress fluids, the total drag coefficient portrays the usual inverse dependence on the Reynolds number and a positive dependence on the Bingham number (Gupta & Chhabra, 2014). It is interesting to note here that the slopes of these plots are more or less similar (for each Reynolds number) at $Bn > 40$. This is presumably due to the fact that the increasing value of Bn signifies a diminishing viscous flow stress $\left(\sim \mu_B U_\infty / D\right)$ rather than a diminishing fluid yield stress. Eventually, at large values of Bn, there is a fully plastic flow regime wherein the drag is governed solely by the value of the fluid yield stress. It is thus possible that $Bn \sim 40$ indicates the onset of the yield-stress-dominated regime, indicated by more or less the same slope seen in Fig. 10. The total drag is not greatly affected by the orientation of the cone in Bingham plastic fluids, though the two values differ from each other by ∼20% at $Re = 100$ and $Bn = 0$ (Table 6). This is, however, due to the self-compensating effect of the variation of the form and viscous drags in opposite directions for the two orientations depending on the contribution of the wake drag and/or the formation of the unsheared regions on the surface of the cone.

Furthermore, Fig. 11 shows the effect of the apex angle on the total drag coefficient for the two extreme values of the Reynolds number (i.e., $Re = 1$ and $Re = 100$) as the apex angle is varied from 20° to 150°. The drag coefficient decreases by almost 60% with this increase in α value; with the bulk of the change occuring between $\alpha = 20^\circ$ and $\alpha = 55^\circ$ in line with previous findings for the creeping-flow regime (Ahonguio et al., 2015). Finally, following the strategy employed in our recent studies (Nirmalkar et al., 2013), the effect

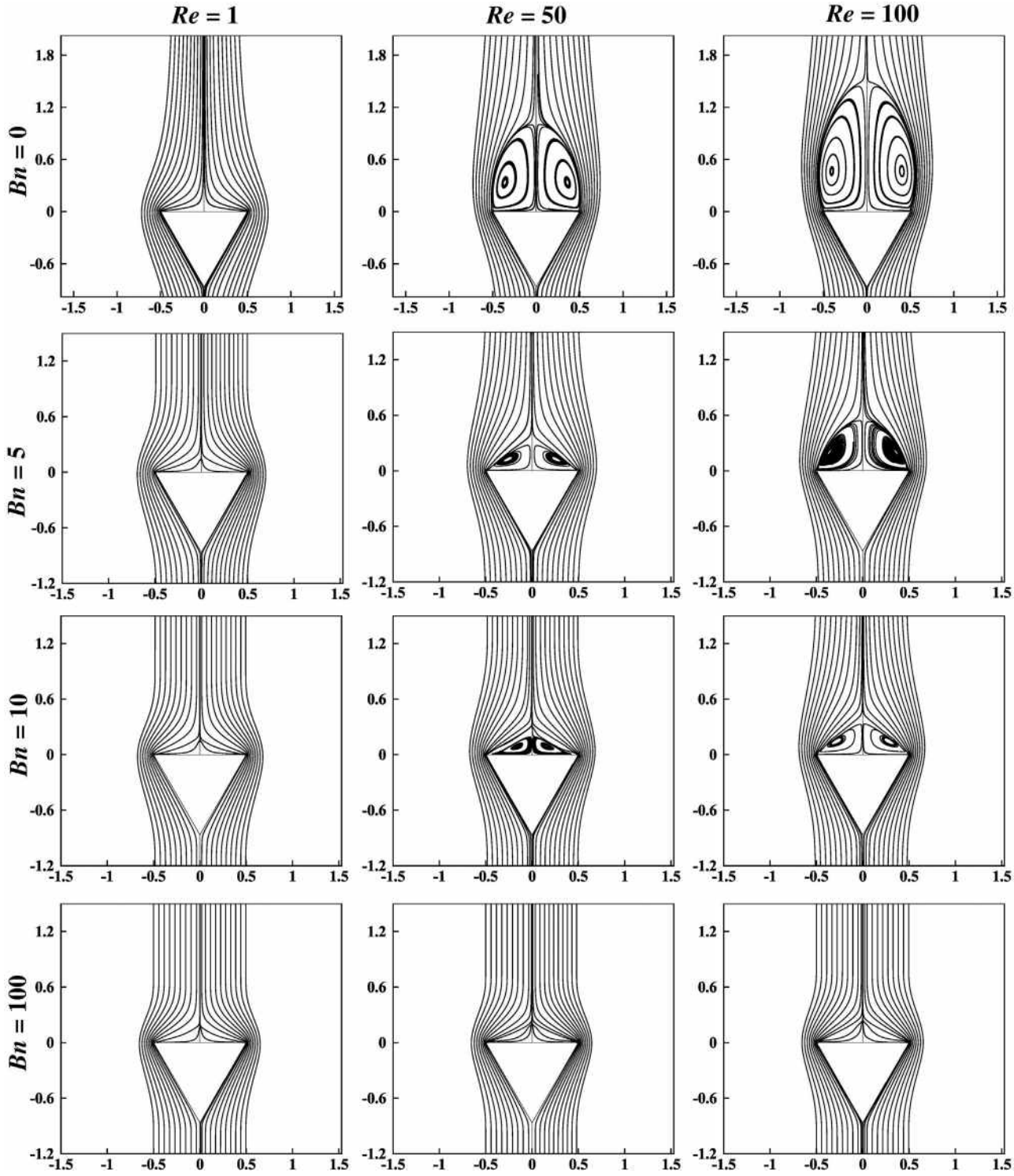


Fig. 6. Effects of the Bingham number and Reynolds number on streamline contours for $\alpha = 60^{\circ}$ (configuration 1).

of the Bingham number on the drag coefficient can be accounted for by redefining the Reynolds number as Re^*:

$$Re^* = \frac{Re}{1 + Bn}. \tag{17}$$

Fig. 12 show the results plotted in this format with the drag coefficient vs. modified Reynolds number, where the results for different values of the Bingham number do collapse onto the same curve except for very small values of the Bingham number. For the sake of comparison, the drag curve for a sphere in Newtonian fluids is also included (Clift et al., 1978). Bearing in mind the logarithmic coordinates, the drag on cones in Bingham plastic fluids is much higher than the drag on spheres; however, this difference decreases with an increase in Reynolds number. Note that the increasing value of Re^* is tantamount to a decreasing value of Bn; i.e., heading towards Newtonian fluid behavior. At large values of α, a cone behaves like a disk and the drag curves for a disk and a sphere superimpose on top of each other for Newtonian fluids (Clift et al., 1978). Meanwhile, the drag on a cone

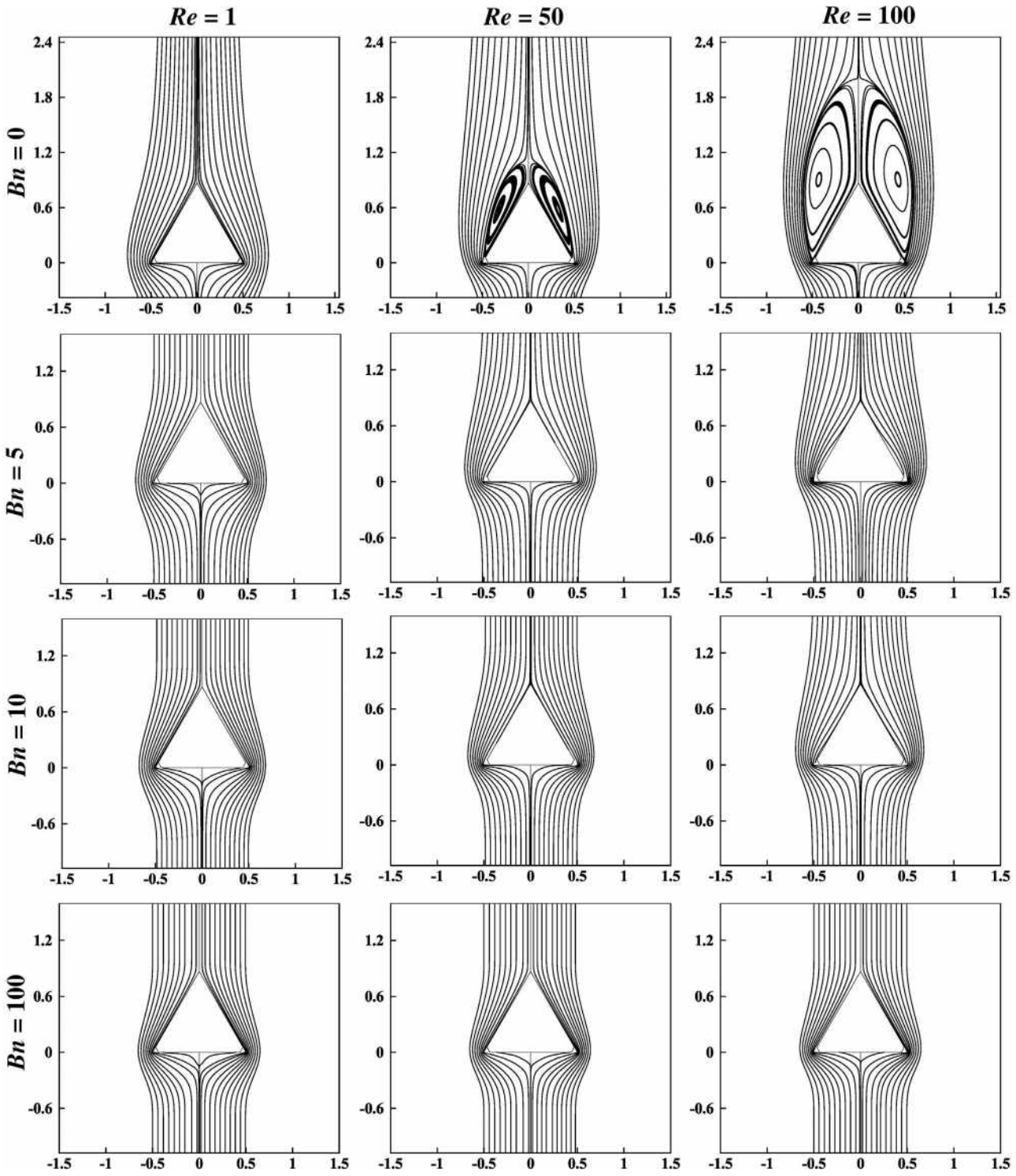


Fig. 7. Effects of the Bingham number and Reynolds number on streamline contours for $\alpha = 60^{\circ}$ (configuration 2).

increases as the value of α decreases; e.g., the data for $\alpha = 20^{\circ}$ are seen to lie at the top in Fig. 12. The present results therefore approach those of a sphere with a gradual increase in α. At this point, the utility of the equal volume sphere diameter was explored to ascertain if the results for different values of cone angle (α) can be further consolidated. This objective could not be achieved. Therefore, a predictive correlation for both orientations (including 70 data points for Newtonian fluids and 250 data points for Bingham plastic fluids) was developed using the present values of the total drag coefficient as a function of the modified Reynolds number, Re^{*}, over a Reynolds number range of $1 \leq Re \leq 100$, Bingham number range of $0 \leq Bn \leq 100$, and cone angle range of $20^{\circ} \leq \alpha \leq 150^{\circ}$:

$$C_{\mathrm{D}} = \frac{64\left(1 + \mathrm{a}Re^{*b}\right)(2 - \cos\ \alpha)^{c}}{\pi Re^{*}}. \tag{18}$$

At the limit $\alpha = 180^{\circ}$, a cone approaches a disk and the form of Eq. (18) approaches this limit (Clift et al., 1978). The numerical

Table 6
Effect of the cone angle on the total drag coefficient (C_D) for configuration 1 and configuration 2.

C_D for configuration 1 (apex downward)								
Bn	$\alpha(^\circ)$							
	20	40	50	55	60	90	120	150
Re = 1								
0	34.036	26.543	25.174	24.714	24.351	23.278	22.917	22.938
1	122.34	91.564	86.045	84.210	82.751	78.510	77.404	77.204
10	548.85	413.00	387.90	379.57	372.83	353.23	348.00	347.56
20	963.57	729.07	685.27	670.54	658.68	623.68	614.81	613.49
50	2140.2	1634.1	1538.1	1509.6	1479.3	1400.5	1380.1	1377.2
100	4034.0	3101.8	2923.8	2863.1	2814.0	2665.9	2627.0	2621.5
Re = 20								
0	3.1325	2.5720	2.4885	2.4659	2.4494	2.4408	2.4841	2.5268
1	6.5040	5.0570	4.8190	4.7121	4.6886	4.5663	4.5850	4.6276
10	27.563	20.837	19.617	19.213	18.900	18.020	17.855	17.869
20	48.262	36.583	34.420	33.768	33.118	31.450	31.079	31.058
50	107.06	81.781	77.000	75.490	74.074	70.188	69.214	69.095
100	201.79	155.14	146.25	143.00	140.78	133.40	131.49	131.22
Re = 50								
0	1.7246	1.4901	1.4691	1.4674	1.4694	1.5197	1.5890	1.6468
1	2.9061	2.3778	2.3084	2.2915	2.2820	2.3081	2.3838	2.4534
10	11.181	8.5770	8.1348	7.9958	7.8878	7.6511	7.6758	7.7474
20	19.425	14.827	14.003	13.734	13.521	12.974	12.926	12.990
50	42.900	32.239	30.957	30.323	29.815	28.354	28.051	28.066
100	80.765	62.142	58.606	57.404	56.434	53.548	52.843	52.779
Re = 100								
0	1.1433	1.0288	1.0284	1.0332	1.0402	1.1032	1.1744	1.2325
1	1.6833	1.4514	1.4358	1.4371	1.4422	1.5136	1.6049	1.6812
10	5.7387	4.5136	4.4512	4.2749	4.2384	4.2033	4.2867	4.3848
20	9.8432	7.6207	7.2497	7.1328	7.0485	6.8841	6.9482	7.0453
50	21.549	16.584	15.678	15.378	15.146	14.533	14.488	14.573
100	40.453	31.191	29.451	28.859	28.394	27.045	26.786	26.822
C_D for configuration 2 (apex upward)								
Bn	$\alpha(^\circ)$							
	20	40	50	55	60	90	120	150
Re = 1								
0	34.116	26.566	25.194	24.734	24.370	23.296	23.013	22.954
1	122.59	91.611	86.087	84.247	82.800	78.566	77.484	77.279
10	550.05	413.48	388.35	379.93	373.36	353.82	348.91	348.12
20	965.56	730.08	686.22	671.50	659.77	624.83	615.9	614.53
50	2146.0	1636.5	1540.6	1508.0	1482.0	1403.7	1383.0	1379.7
100	4048.3	3107.3	2928.9	2868.0	2819.4	2672.5	2633.4	2627.0
Re = 20								
0	3.2193	2.6738	2.5935	2.5705	2.5546	2.5310	2.5450	2.5570
1	6.5752	5.1426	4.9111	4.8395	4.7871	4.6659	4.6599	4.6614
10	27.630	20.919	19.701	19.297	18.987	18.110	17.937	17.930
20	48.320	36.658	34.501	33.782	33.210	31.544	31.167	31.133
50	107.24	81.886	77.119	75.501	74.208	70.344	69.356	69.226
100	202.26	155.39	146.48	143.44	141.02	133.71	131.79	131.50
Re = 50								
0	1.8156	1.6326	1.6280	1.6315	1.6370	1.6739	1.6944	1.6986
1	2.9990	2.5178	2.4672	2.4575	2.4533	2.4552	2.4862	2.5062
10	11.297	8.7222	8.2897	8.1543	8.0547	7.8337	7.8254	7.8243
20	19.510	14.959	14.144	13.881	13.677	13.142	13.078	13.078
50	42.975	32.921	31.055	30.427	29.928	28.485	28.173	28.157
100	80.934	62.238	58.702	57.501	56.543	53.690	52.983	52.903
Re = 100								
0	1.2468	1.2364	1.2670	1.2788	1.2884	1.3106	1.3059	1.2945
1	1.7893	1.6384	1.6412	1.6490	1.6606	1.7254	1.7498	1.7533
10	5.8802	4.7028	4.5481	4.5100	4.4855	4.4578	4.4562	4.4575
20	9.9887	7.8088	7.4564	7.3500	7.2777	7.1449	7.1441	7.1428
50	21.659	16.731	15.848	15.559	15.332	14.749	14.681	14.686
100	40.576	31.285	29.561	28.982	28.522	27.205	26.939	26.930

Table 7
Values of fitted constants of Eq. (18).

Cone angle	Fluid	*a*	*b*	*c*	% error	
					Mean	Maximum
$\left(20^o \le \alpha \le 150^o\right)$	Newtonian fluids (*Bn* = 0)	0.47	0.51	−0.33	8	20
	Bingham plastic fluids (*Bn* > 0)	1.13	0.12	−0.38	13	33

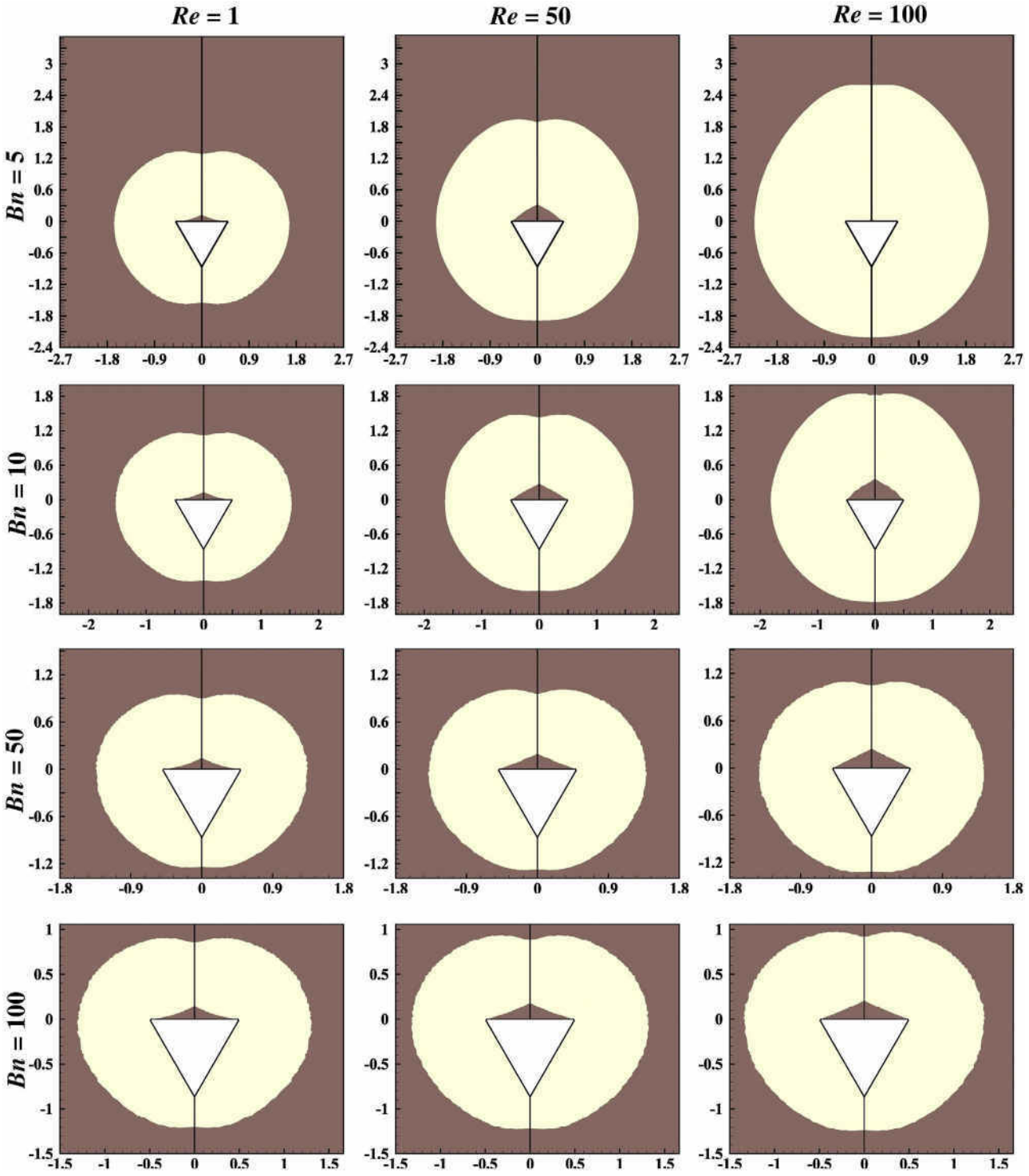


Fig. 8. Effects of the Bingham number and Reynolds number on yield surfaces (configuration 1) for $\alpha = 60^{\circ}$.

constants are listed in Table 7. The resulting mean and maximum deviations are respectively ~8% and ~20% for Newtonian fluids and ~13% and ~33% for Bingham plastic fluids. Only 20 data points (for Bingham plastic fluids) in each orientation exhibit deviations >25% from the values predicted with Eq. (18). The deviations for both orientations were maximum at the lowest values of the Bingham number, $Bn = 1$, and/or at the highest value of the Reynolds number, $Re = 100$. Statistical analysis of the results shows that deviation between the predicted (from Eq. (18)) and numerical values decreases with increasing Bingham number and increases with increasing Reynolds number. Further attempts to improve the degree of fit proved to be unsuccessful even after introducing more fitting parameters. While the exact reasons for such large deviations are not immediately obvious, the deviations are comparable to those reported for drag results of spheres and cylinders (Chhabra, 2006; Nirmalkar et al., 2013) and other shapes (Rajitha et al., 2006). For the sake of comparison, results for a sphere in Newtonian fluid (Clift et al., 1978) are also included here. A cursory examination of these results shows that the drag of a cone in Bingham plastic fluids follows the same trend as a sphere in Newtonian fluids.

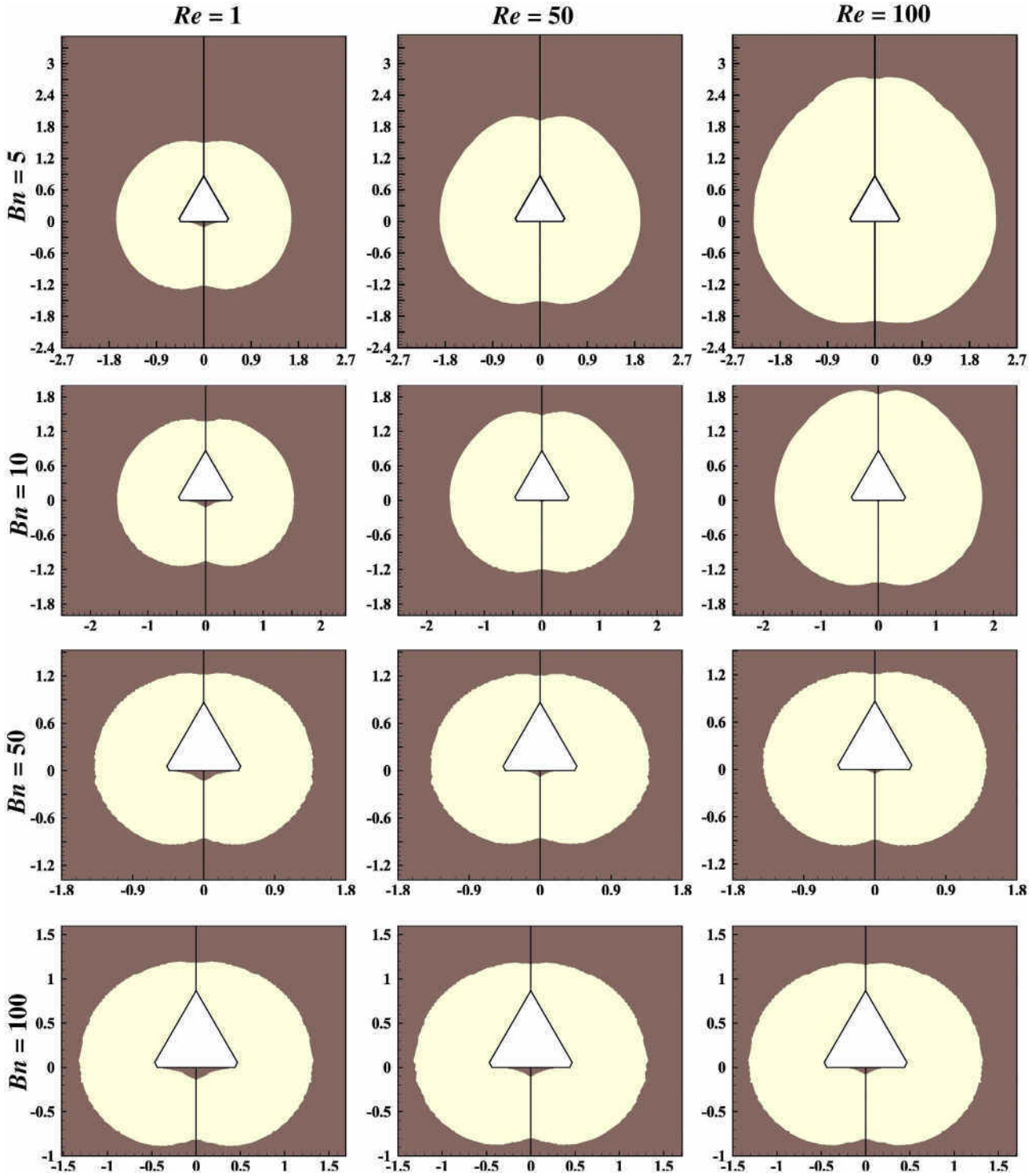


Fig. 9. Effects of the Bingham number and Reynolds number on yield surfaces (configuration 2) for $\alpha = 60^{\circ}$.

Conclusions

The effect of the orientation of a cone with varying apex angle on the drag of the cone in Bingham plastic fluid was investigated for $1 \leq Re \leq 100$, $20^{\circ} \leq \alpha \leq 150^{\circ}$, and $0 \leq Bn \leq 100$. Extensive numerical results were analyzed in terms of streamline contours, the morphology of yielded/unyielded regions, and the drag coefficients. It was found that, even at high Reynolds numbers, fluid deformation reduced at large values of the Bingham number, and the tendency for wake formation was thus also suppressed. In other words, the fluid yield stress imparts stability to the flow near the falling cone. The drag coefficient has a positive dependence on the Bingham number. Over the range of the Reynolds numbers investigated in the present work, the orientation exerts little effect on the overall hydrodynamic drag. This is possibly due to the fact that the cone was held fixed in the present study and it may not be so for a freely falling cone. Finally, the present numerical predictions are in line with scant experimental results for Newtonian fluids.

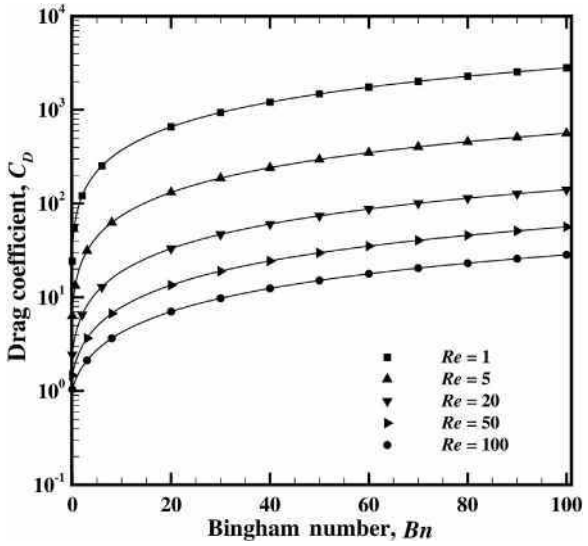


Fig. 10. Variations of the drag coefficient with the Bingham number at different Reynolds numbers at $\alpha = 60^\circ$ for configuration 1 (symbols) and configuration 2 (lines).

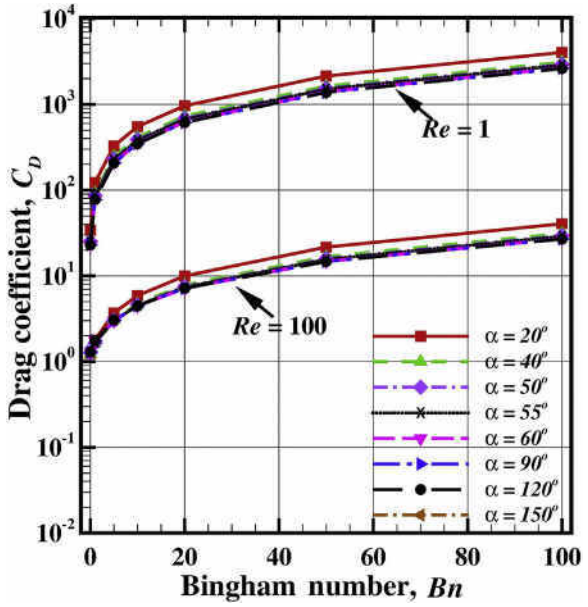


Fig. 11. Variations of drag coefficient with Bingham number at different cone angles and two extreme values of Reynolds number for configuration 1 (symbols) and configuration 2 (lines).

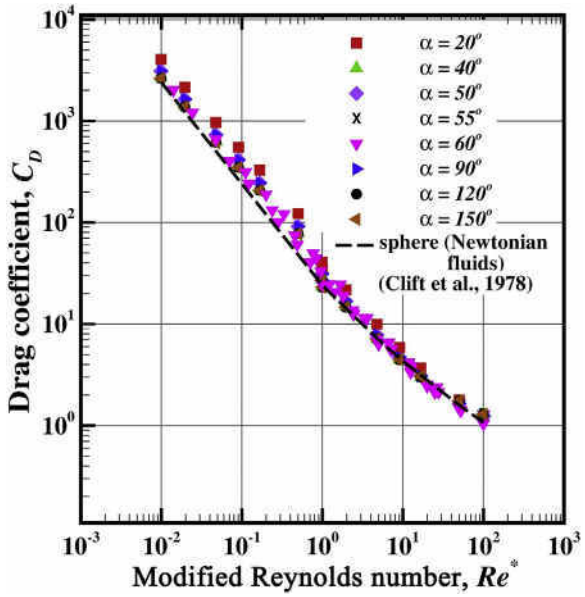


Fig. 12. Variations of the drag coefficient with the modified Reynolds number at different α values for configuration 1 (filled symbols) and configuration 2 (open symbols). The broken line corresponds to a sphere settling in a Newtonian fluid.

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